

SAFETY DATA SHEET**1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION**

Product Name	Penguin Seal 353White
Application of Product	For industry use.
Manufacturer Name	Sunstar Chemical (Thailand) Co.,Ltd.
Address	624 Moo 4 Bangpoo Industrial Estate Soi 7A Sukhumvit Rd., Praksa, Muang Samutprakarn, Samutprakarn 10280 Thailand
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2. HAZARDS IDENTIFICATION**PHYSICAL HAZARDS**

Flammable solids

Category 2

HEALTH HAZARDS

Acute toxicity (Oral)

Not classified

Acute toxicity (Dermal)

Not classified

Acute toxicity (Inhalation: Gases)

Not applicable

Acute toxicity: (Inhalation: Vapours)

Category 4

Acute toxicity (Inhalation: Dusts and mists)

Classification not possible

Skin corrosion/irritation

Category 2

Serious eye damage/eye irritation

Category 2

Respiratory sensitization

Category 1

Skin sensitization

Category 1

Germ cell mutagenicity

Not classified

Carcinogenicity

Category 2

Reproductive toxicity

Category 1A

Specific target organ toxicity - Single exposure

Category 1 (Central nervous system, Respiratory
organs, Systemic toxicity)

Specific target organ toxicity - Repeated exposure

Category 1 (Central nervous system, Kidney,
Respiratory organs)

Aspiration hazard

Category 1

ENVIRONMENTAL HAZARDS

Hazardous to the aquatic environment (Acute)

Category 2

Hazardous to the aquatic environment (Long-term)

Category 3

Hazardous to the ozone layer

Classification not possible

ALLOCATION OF LABEL ELEMENTS**Hazard symbol****Signal word**

Danger

Hazard statement

H228 Flammable solid.

H304 May be fatal if swallowed and enters airways.

H315 Causes skin irritation.

H317 May cause an allergic skin reaction.

H319 Causes serious eye irritation.

H332 Harmful if inhaled.

H334 May cause allergy or asthma symptoms or breathing difficulties if inhaled.

H351 Suspected of causing cancer.

H360 May damage fertility or the unborn child

H370 Causes damage to organs (Central nervous system, Respiratory organs, Systemic toxicity).

H372 Causes damage to organs (Central nervous system, Kidneys, Respiratory organs) through prolonged
or repeated exposure.

H401 Toxic to aquatic life.

H412 Harmful to aquatic life with long lasting effects.

Precautionary statements**Prevention**

- P203 Obtain, read and follow all safety instructions before use.
 P210 Keep away from heat/sparks/open flames/hot surfaces. No smoking.
 P233 Keep container tightly closed.
 P240 Ground/bond container and receiving equipment.
 P241 Use explosion-proof electrical/ventilating/lighting/ equipment.
 P242 Use only non-sparking tools.
 P243 Take action to prevent static discharges.
 P260 Do not breathe dust/fume/gas/mist/vapours/spray.
 P261 Avoid breathing dust/fume/gas/mist/vapours/spray.
 P264 Wash thoroughly after handling.
 P264+P265 Wash hands thoroughly after handling. Do not touch eyes.
 P270 Do not eat, drink or smoke when using this product.
 P271 Use only outdoors or with adequate ventilation.
 P272 Contaminated work clothing should not be allowed out of the workplace.
 P273 Avoid release to the environment.
 P280 Wear protective gloves/protective clothing/eye protection/face protection.
 P284 In case of inadequate ventilation wear respiratory protection.

Response

- P301+P316 IF SWALLOWED: Get emergency medical help immediately.
 P302+P352 IF ON SKIN: Wash with plenty of soap and water.
 P303+P361+P353 IF ON SKIN or hair: Take off immediately all contaminated clothing.
 Rinse affected areas with water or shower.
 P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
 P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses,
 if present and easy to do. Continue rinsing.
 P308+P316 IF exposed or concerned: Get emergency medical help immediately.
 P317 Get medical help.
 P318 IF exposed or concerned, get medical advice.
 P319 Get medical help if you feel unwell.
 P321 Specific treatment (see label).
 P331 Do NOT induce vomiting.
 P332+P317 If skin irritation occurs: Get medical help.
 P333+P317 If skin irritation or rash occurs: Get medical help.
 P337+P317 If eye irritation persists: Get medical help.
 P342+P316 If experiencing respiratory symptoms: Get emergency medical help immediately.
 P362+P364 Take off contaminated clothing and wash it before reuse.
 P370+P378 In case of fire: Use carbon dioxide to extinguish.

Storage

- P403 Store in a well-ventilated place.
 P403+P235 Store in a well-ventilated place. Keep cool.
 P405 Store locked up.

Disposal

- P501 Dispose of contents/container to..... (in accordance with local/regional/national/
 international regulation).

3. COMPOSITION/INFORMATION ON INGREDIENTS**Substance/Preparation****Preparation****General Name**

Polychloroprene rubber type sealant

Components Contributing to the Hazard

Components	Cas No	%WT
Toluene	108-88-3	39
Titanium dioxide	13463-67-7	≤ 5
2,2'-Methylenebis(4-methyl-6-tert-butylphenol)	119-47-1	≤ 1
Zinc oxide	1314-13-2	≤ 1

4. FIRST AND MEASURES

Inhalation

Remove to fresh air and keep at rest in a position comfortable for breathing.
If experiencing respiratory symptoms, get emergency medical help immediately.

Skin Contact

Wash with plenty of soap and water.
If contact on skin or hair, take off immediately all contaminated clothing. Rinse affected areas with water or shower.
If skin irritation or rash occurs, get emergency medical help immediately.

Eye Contact

Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
If eye irritation persists, get medical advice/attention.

Ingestion

Do not induce vomiting.
If you feel unwell, get medical attention immediately.

5. FIRE-FIGHTING

Extinguishing Media

Carbon dioxide, dry chemical, foam

Special Hazards

Toxic gases (chlorine compound) will foam upon combustion.

Special Methods

Evacuate non essential personnel.
Move container from fire areas if it can be done without risk.
Keep personnel removed from and upwind of fire.

Protection of Firefighters

Wear Self-Contained Breathing Apparatus (SCBA).

6. ACCIDENTAL RELEASE MEASURES

Personal Precautions

Wear protective equipment and avoid contact with skin.

Environmental Precautions

Do not allow to enter sewers/surface or ground water.

Methods for Cleaning up

Absorb with liquid-binding materials (sand, sawdust etc.).
Scoop material into containers for disposal.

7. HANDLING AND STORAGE

Handling

Technical Measures

Obtain, read and follow all safety instructions before use.
Ground/bond container and receiving equipment.
Use explosion-proof electrical/ventilating/lighting/ equipment.
Use only non-sparking tools.
Take action to prevent static discharges.

Precautions

Do not breathe dust/fume/gas/mist/vapours/spray.
Wash hands thoroughly after handling. Do not touch eyes.
Do not eat, drink or smoke when using this product.
Use only outdoors or with adequate ventilation.
Contaminated work clothing should not be allowed out of the workplace.
Avoid release to the environment.
Wear protective gloves/protective clothing/eye protection/face protection.
In case of inadequate ventilation wear respiratory protection.

Storage

Suitable Storage Conditions

Keep away from heat/sparks/open flames/hot surfaces. No smoking.
Keep container tightly closed.
Store in a cool and well ventilated place.
Protect material from direct sunlight.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Components	ACGIH (2024,TWA)
Toluene	20 ppm
Titanium dioxide	2.5 mg/m ³
2,2'-Methylenebis(4-methyl-6-tert-butylphenol)	No data available
Zinc oxide	2 mg/m ³

Personal Protective Equipment

Hand Protection

Protective gloves

Eye Protection

Safety glasses

Skin and Body Protection

Avoid contact with skin.

Hygiene Measures

Wash hands before taking a break and at the end of work.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical State

Appearance	White paste
Odour	Solvent odor
pH	Not established
Melting point/freezing point	Not applicable
Initial boiling point and boiling range	Not applicable
Flashpoint	Not available
Evaporation rate	Not available
Flammability	Not available
Upper/lower flammability or explosive limits	Not available
Vapour pressure	Not available
Relative density	About 1.2
Solubility	Not available
Partition coefficient:n-octanol/water	Not available
Auto-ignition temperature	Not available
Viscosity	250 - 350 Pa.s at 20°C

10. STABILITY AND REACTIVITY

Stability

Stable under normal conditions.

Possible Hazardous Reactions

Reactive with water and amine compound

Conditions to Avoid

Avoid water and amine compound

Hazardous Decomposition Products

Chlorine compound (in addition to CO , CO₂ , H₂O)

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Toluene :	LD50 (Oral) :	Rats	5,000 mg/kg	(Ministry of the Environment, 2002)
			5,580 mg/kg	(EU-RAR, 2003)
			5,900 mg/kg, 6,000 mg/kg,	(EHC 52, 1985)
			and 7,530 mg/kg	
			7,000 mg/kg	(JECFA 518, 1981)
			7,300 mg/kg	(ATSDR, 2000)
	LD50 (Dermal) :	Rats	12,000 mg/kg	(ACGIH, 2007)
			14,100 mg/kg	(ACGIH, 2007)
		Rabbits	12,400 mg/kg	(EU-RAR, 2003)
	LC50 (Inhalation dusts and mists(4H))	Rats	7,460 ppm, 3,319-7,646 ppm,	(EU-RAR, 2003)
			and 8,762 ppm	
			4,000 ppm, 8,000 ppm, and	(PATTY 5th, 2001)
			8,800 ppm	

Titanium dioxide : LD50 (Oral) :	Rats	>2,000 mg/kg and >5,000 mg/kg (SIDS, 2015)	
		>10,000 mg/kg	(HSDB, 2016 and Ministry of the Environment, 2010)
		>12,000 mg/kg and >20,000 mg/kg	(Ministry of the Environment, 2010)
LD50 (Dermal) :	Hamsters	>10,000 mg/kg	(HSDB, 2016 and Ministry of the Environment, 2010)
LC50 (Inhalation dusts and mists(4H))	Rats	>5.09 mg/L and >3.43 mg/L	(SIDS, 2015)
2,2'-Methylenebis(4-methyl-6-tert-butylphenol) LD50 (Oral) :	Rats	>5,000 mg/kg	(SIAR, 2001 and REACH, 2022)
		>10,000 mg/kg	(SIAR, 2001)
		>10,000 mg/kg	(REACH, 2022)
Zinc oxide :	LD50 (Dermal) :	Rabbits	>10,000 mg/kg (EU-RAR, 2004)
	LD50 (Oral) :	Rats	>5,000 mg/kg and >15,000 mg/l (EPA Pesticide, 1992)
	LD50 (Dermal) :	Rabbits	>5,000 mg/kg (EU-RAR, 2004)
	LC50 (Inhalation dusts and mists(4H))	Rats	>5.7 mg/L (EU-RAR, 2004)

Skin corrosion/irritation

Toluene :	Category 2
Titanium dioxide :	Not classified
2,2'-Methylenebis(4-methyl-6-tert-butylpl	Not classified
Zinc oxide :	Not classified

Serious eye damage/eye irritation

Toluene :	Category 2B
2,2'-Methylenebis(4-methyl-6-tert-butylpl	Not classified
Zinc oxide :	Not classified

Skin sensitization

Toluene :	Not classified
Titanium dioxide :	Not classified
2,2'-Methylenebis(4-methyl-6-tert-butylpl	Not classified
Zinc oxide :	Not classified

Germ cell mutagenicity

Toluene :	Not classified
2,2'-Methylenebis(4-methyl-6-tert-butylpl	Not classified

Carcinogenicity

Titanium dioxide :	Category 2
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Reproductive toxicity

Toluene :	Category 1A
2,2'-Methylenebis(4-methyl-6-tert-butylpl	Category 1B
Zinc oxide :	Category 2

Specific target organ toxicity - Single exposure

Toluene :	Category 1 (Central nervous system)
	Category 3 (Respiratory tract irritation, Narcotic effects)
Zinc oxide :	Category 1 (Respiratory system, Systemic toxicity)

Specific target organ toxicity - Repeated exposure

Toluene :	Category 1 (Central nervous system, Kidney)
Titanium dioxide :	Category 1 (Respiratory organs)
2,2'-Methylenebis(4-methyl-6-tert-butylpl	Category 1 (Respiratory organs (male))

Aspiration hazard

Toluene :	Category 1
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12. ECOLOGICAL INFORMATION

Hazardous to the aquatic environment (Acute)

Toluene :	EC50 (48H) :	Crustacea (Ceriodaphnia dubia)	3.78 mg/L	(NITE,CERI,NEDO, 2006)
2,2'-Methylenebis(4-methyl-6-tert-butylphenol)	EC50 (48H) :	Crustacea (Daphnia magna)	>0.01 mg/L	(Ministry of the Environment, 2005)
Zinc oxide :	LC50 (48H) :	Crustacea (Daphnia magna)	0.098 mg/L	(NITE,CERI,NEDO, 2008)

Hazardous to the aquatic environment (Long-term)

Toluene :	NOEC (7days) :	Crustacea (Ceriodaphnia dubia)	0.74 mg/L	(NITE,CERI,NEDO, 2006)
2,2'-Methylenebis(4-methyl-6-tert-butylphenol)	NOEC (7days) :	Crustacea (Daphnia magna)	>0.0057 mg/L	(Ministry of the Environment, 2005)
Zinc oxide :	LC50 (48H) :	Crustacea (Daphnia magna)	0.098 mg/L	(NITE,CERI,NEDO, 2008)
	NOEC (72H) :	Algae (Pseudokirchneriella subcapitata)	0.024 mg/L	(EU-RAR, 2010)

13. DISPOSAL CONSIDERATIONS

Waste from Residues

Follow the all regulations in your country.

Contaminated Packaging

Follow the all regulations in your country.

14. TRANSPORT INFORMATION

UN Class 4.1

UN No 1325

Packing group II

Specific Precautionary Transport Measures and Conditions

Follow the all regulations in your country.

15. REGULATORY INFORMATION

Follow the all regulations in your country.

16. OTHER INFORMATION

The information herein is given in good faith, but no warranty, express or implied, is made.

The information presented herein is believed to be accurate but is not warranted. Recipients are advised to confirm in advance that the information is current, applicable and suitable to there circumstances.